Soal 2 :

Tentukan $q(t)$ bila pada $t=0$, $q=10$, $q'=0$ dan sketsa grafik dari soal:

1. $q'' + 4q' + 3q = 75$

Persamaan karakteristik : $m^2 + 4m + 3 = 0$

diperoleh akar – akarnya : $(m+1)(m+3)$

$m_1 = -1$ V $m_2 = -3$

Karena $q(t) = 75$, $q = c$, $q' = 0$, dan $q'' = 0$ substitutikan pada PD:

$q(t) = q'' + 4q' + 3q$

$75 = 0 + 4(0) + 3C$

$75 = 3C$

$C = 25$

∴ Penyelesaian umumnya adalah $q(t) = C_1 e^{-t} + C_2 e^{-3t} + 25$

Jika $t=0$, $q=10$, $q'=0$

maka, $q(0) = C_1 e^{-(0)} + C_2 e^{-3(0)} + 25$

$10 = C_1 + C_2 + 25$

$10 - 25 = C_1 + C_2$

$-15 = C_1 + C_2$ ..... (1)

$q'(t) = -C_1 e^{-t} - 3C_2^{-3t}$

$q'(0) = -C_1 e^{-(0)} - 3C_2^{-3(0)}$

$0 = -C_1 - 3C_2$ ..... (2)

Metode eliminasi :

(1) $-15 = C_1 + C_2$

(2) $0 = -C_1 - 3C_2$ –

$15 = -C_1 - C_2$

$0 = -C_1 - 3C_2$ –

$15 = 2C_2$

$C_2 = \frac{15}{2}$

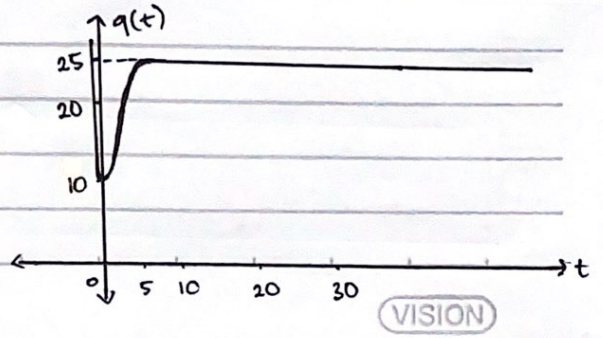
$-15 = C_1 + \frac{15}{2}$

$-15 - \frac{15}{2} = C_1$

$-\frac{45}{2} = C_1$

∴ Penyelesaian khususnya adalah

$q(t) = -\frac{45}{2} e^{-t} + \frac{15}{2} e^{-3t} + 25$



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2. $q'' + 4q' + 4q = 75$

Persamaan karakteristik = $m^2 + 4m + 4 = 0$

diperoleh akar - akarnya = $(m + 2)(m + 2)$

$m_1 = m_2 = -2$

Karena $q(t) = 75$, $q = c$, $q' = 0$, $q'' = 0$ substitusikan pada PD:

$q(t) = q'' + 4q' + 4q$

$75 = 0 + 4(0) + 4c$

$75 = 4C$

$\frac{75}{4} = C$

∴ Penyelesaian umumnya adalah $q(t) = e^{-2t}(C_1 + C_2 t) + \frac{75}{4}$

Jika $t = 0$, $q = 10$, $q' = 0$

maka , •) $q(0) = e^{-2(0)}(C_1 + C_2(0)) + \frac{75}{4}$

$10 = C_1 + \frac{75}{4}$

$C_1 = -\frac{35}{4}$ ..... (1)

•) $q'(t) = -2C_1 e^{-2t} - 2C_2 t e^{-2t} + C_2 e^{-2t}$

$q'(0) = -2C_1 e^{-2(0)} - 2C_2(0) e^{-2(0)} + C_2 e^{-2(0)}$

$0 = -2C_1 + C_2$ ..... (2)

Metode substitusi :

$C_1 = -\frac{35}{4}$

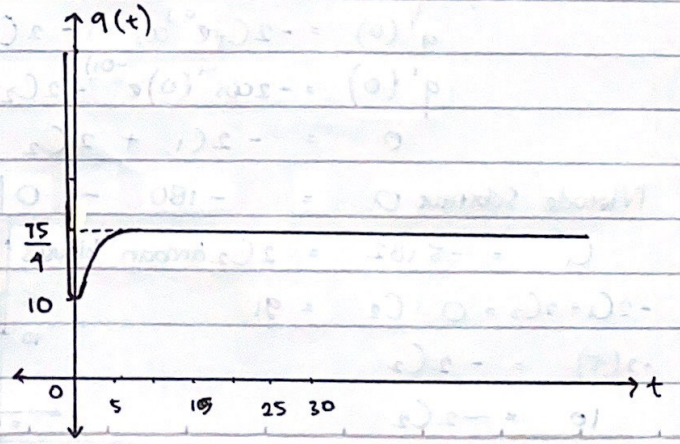
$-2C_1 + C_2 = 0$

$-2\left(-\frac{35}{4}\right) + C_2 = 0$

$C_2 = -\frac{70}{4}$

$C_2 = -\frac{35}{2}$

∴ Penyelesaian khususnya adalah $q(t) = e^{-2t}\left(-\frac{35}{4} - \frac{35}{2}t\right) + \frac{75}{4}$



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3. $q'' + 4q' + 5q = 75$

Persamaan karakteristik : $m^2 + 4m + 5 = 0$

diperoleh akar - akarnya $= \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$, dimana $a = 1$, $b = 4$, $c = 5$

$$= \frac{-4 \pm \sqrt{4^2 - 4(1)(5)}}{2(1)}$$

$$= \frac{-4 \pm \sqrt{16 - 20}}{2}$$

$$= \frac{-4 \pm \sqrt{-4}}{2}$$

$$= -2 \pm j2 \rightarrow m_{1,2}$$

Karena $q(t) = 75$, $q'' = 0$, $q' = 0$, $q = c$, Substitusikan pada PD :

$q(t) = q'' + 4q' + 5q$

$75 = 0 + 4(0) + 5C$

$75 = 5C$

$C = 15$

∴ Persamaan umumnya adalah

$q(t) = e^{-2t}(C_1 \cos 2t + C_2 \sin 2t) + 15$

Jika $t = 0$, $q = 10$, $q' = 0$

maka, ·) $q(0) = e^{-2(0)}(C_1 \cos 2(0) + C_2 \sin 2(0)) + 15$

$10 = C_1 + 15$

$C_1 = 10 - 15$

$C_1 = -5$ ..... (1)

·) $q'(t) = -2C_1 e^{-2t}\cos(2t) - 2C_2 e^{-2t}\sin(2t) - 2C_1 e^{-2t}\sin(2t) + 2C_2 e^{-2t}\cos(2t)$

$q'(0) = -2C_1 e^{-2(0)}\cos(0) - 2C_2 e^{-2(0)}\sin(0) - 2C_1 e^{-2(0)}\sin(0) + 2C_2 e^{-2(0)}\cos(0)$

$0 = -2C_1 - 0 - 0 + 2C_2$

$0 = -2C_1 + 2C_2$ ..... (2)

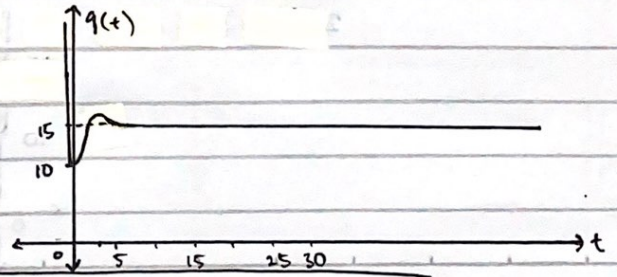
Metode Substitusi :

$C_1 = -5$

$-2C_1 + 2C_2 = 0$

$-2(-5) = -2C_2$

$C_2 = -5$



∴ Persamaan khususnya adalah $q(t) = e^{-2t}(-5\cos 2t - 5\sin 2t) + 15$

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